



Population decline of amphipod *Monoporeia affinis* in Northern Europe: consequence of food shortage and competition?

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ABSTRACT

In the present study, we investigate the effects of food quality (with three distinct algal diets) and quantity (high vs. low carbon and nitrogen contents) and competition (with natural polychaete competitor, *Marenzelleria* spp) on the semelparous amphipod *Monoporeia affinis* fitness and reproduction. Contrary to other studies on this species, our results showed that amphipod females have the ability to adjust their offspring investment when conditions were significantly altered before mating in terms of food quantity, while food quality was of less importance. In fact, there was only a low beneficial effect on amphipod reproduction and embryonic development of a presumed high quality diatom diet. Also, the presence of a natural competitor did not affect amphipod reproductive outcome. The results are viewed in the context of a dramatic decline observed on the amphipod populations in the Gulf of Bothnia. Though, several plausible hypotheses have been suggested for the decline, including competition, new anthropogenic substances, hypoxia and predation, we hypothesize that the population decline is related to food shortage and malnutrition due to an increased percentage low quality terrestrial carbon (rich in humus) from river outflow into the Gulf of Bothnia. Concomitantly, the bacterial food web was enhanced, which has a lower trophic efficiency. We also suggest that a secondary effect of malnutrition is an increased susceptibility to other types of stressors such as contaminants and parasites.

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1. Introduction

Semelparity is considered as an optimal life strategy in a constant environment, while iteroparity is selected in stochastic environments (Gurney and Middleton, 1996; Katsukawa et al., 2002), since environmental fluctuations make it necessary to “diversify the risk of reproduction” (Katsukawa et al., 2002). The risks associated with a semelparous reproductive strategy are due to the extreme reproductive trade-offs which have to be made since resources are allocated for reproduction at the expense of somatic functions when breeding, (Morbey and Abrams, 2004). In many semelparous species, like mayflies, the breeding lifespan is very short, (hours-days). The semelparous Baltic amphipod *Monoporeia affinis* diverges by having a long reproductive cycle. Its oogenesis starts in late summer, mating takes place in November while the gravid females carry their offspring for more than two months. The age at reproduction varies between 1–4 years the Baltic Sea. There have been several theories regarding what governs the life span of *M. affinis*. Segerstråle (1950) suggested differences in growth rate caused by temperature, Siegfried (1985) proposed that the growth rate was mainly affected by environmental productivity, while Leonardsson et al. (1988) showed that it was not

the growth rate per se, but rather the size of the amphipods that was important. He suggested that these amphipods need to reach a threshold size in order to reproduce. This theory was strengthened by Hill et al. (1992) whom showed that both length and weight are positively correlated to lipid content and suggest that *M. affinis* needs a lipid content of at least 20% to start reproduction.

To achieve this high lipid content *M. affinis* preferably feeds on newly settled phytoplankton (Aljetlawi et al., 2000; Goedkoop and Johnson, 2001). Despite a small fraction of algae in the gut, Johnson and Wiederholm (1992) reported a strong correlation between yearly growth and spring bloom biovolume. Thus, sedimentated phytoplankton is supposed to be the most important food source, despite its minor fraction in the gut. The spring bloom in the Baltic Sea has normally been dominated by diatoms, which are regarded as high quality food resources compared to bacteria and other species of algae (Volkman et al., 1989; Ederington et al., 1995). They are rich in essential polyunsaturated fatty acids (PUFA) of the linolenic (n-3) and linoleic (n-6) families and the ratio between the linolenic (n-3) and linoleic (n-6) families is regarded as a measure of food quality, which increases with increasing ratio (Goedkoop et al., 2000). Field surveys from 1996 to 2002 in the ultraoligotrophic Lake Vättern and the more eutrophic Lake Vänern demonstrated that the n-3/n-6 ratio in *M. affinis* from Lake Vänern was twice the ratio in Lake Vättern (Sundelin et al., 2008b). It has been observed that female crustaceans

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richer in lipids produce a larger number of eggs (Glazier, 2000). Thus, inter-annual variations in the composition and magnitude of plankton blooms will affect the dietary concentrations of fatty acids (FA) and subsequently strongly affect the growth and reproductive outcome in consumer animal populations (Goedkoop and Johnson, 2001; Garrido et al., 2007, 2008).

Food intake during the autumn has previously been suggested to be of minor importance to gravid females but a study of Aljetjawi and Leonardsson (2003) and own observations emphasized the importance of food availability even during this late stage of reproduction in order to become successful. Between periods of phytoplankton bloom the amphipods can subsist on low quality food, which is facilitated by the low temperatures in their soft bottom habitat, but in March when sediments are deficient in high quality food the amphipods starve and utilise almost exclusively stored lipids as energy source (Lehtonen, 1994). The general theory is that during starvation, larger individuals are superior to smaller specimens due to smaller metabolic costs on a mass basis (Calder, 1984; Nilsson, 1974). This means that we can expect juveniles to be more susceptible to food deficiency than adults, but Aljetjawi and Leonardsson (2003) suggested that in species allocating large amount of fat into reproductive tissue the situation could be reversed. During *M. affinis* sexual maturation an important part of the resources is allocated to reproduction through oogenesis resulting in a minor part of food resources left for maintenance. Experimental studies confirmed their hypothesis since juvenile *M. affinis* showed lower mortality and larger ability to use fresh food resources than adults during starvation (Aljetjawi and Leonardsson, 2002, 2003).

The importance of food for maintenance of the population size is indisputable but other limiting factors, e.g., predation, diseases and environmental stresses, can influence the relative importance of food availability for different species. The very high lipid content in *M. affinis* indicates food availability to be of particular significance for changes in population size and growth. Normally the *M. affinis* populations show cyclic fluctuations in the Gulf of Bothnia covering 6–7 years. However, in 1999 a dramatic decline started on coastal stations in the Gulf of Bothnia and during 2002 and 2003 the decline was extended to the southern and central stations and principally all stations were affected (Karlsson and Leonardsson, 2005). As far as we know, "this pattern of simultaneously low abundances on all coastal and offshore stations has never been seen before" (Aljetjawi, 2003). A similar large-scale decline has been observed for the very closely related amphipod *Diporeia* sp in the Great Lakes in North America where populations declined by 90% from 1993 to 2002 (Nalepa et al., 2006). Both used to be the dominating soft bottoms species. In the Baltic proper, declines of both *M. affinis* and the marine congeneric species *Pontoporeia femorata* have been observed since the late 1970s but contrary to the sudden and dramatic decline in the Gulf of Bothnia this population decline has been a slow and continuous process, interrupted by recovery some years (monitoring data SHARK database SMHI, Sweden). Today the abundance is about 1/10 what it was 35 years ago. The Baltic proper decline is strongly influenced by hypoxia (Eriksson Wiklund and Sundelin, 2004; Cederwall, 2004) and the hypoxia-resistant bivalve *Macoma balthica* (monitoring data SHARK database SMHI, Sweden).

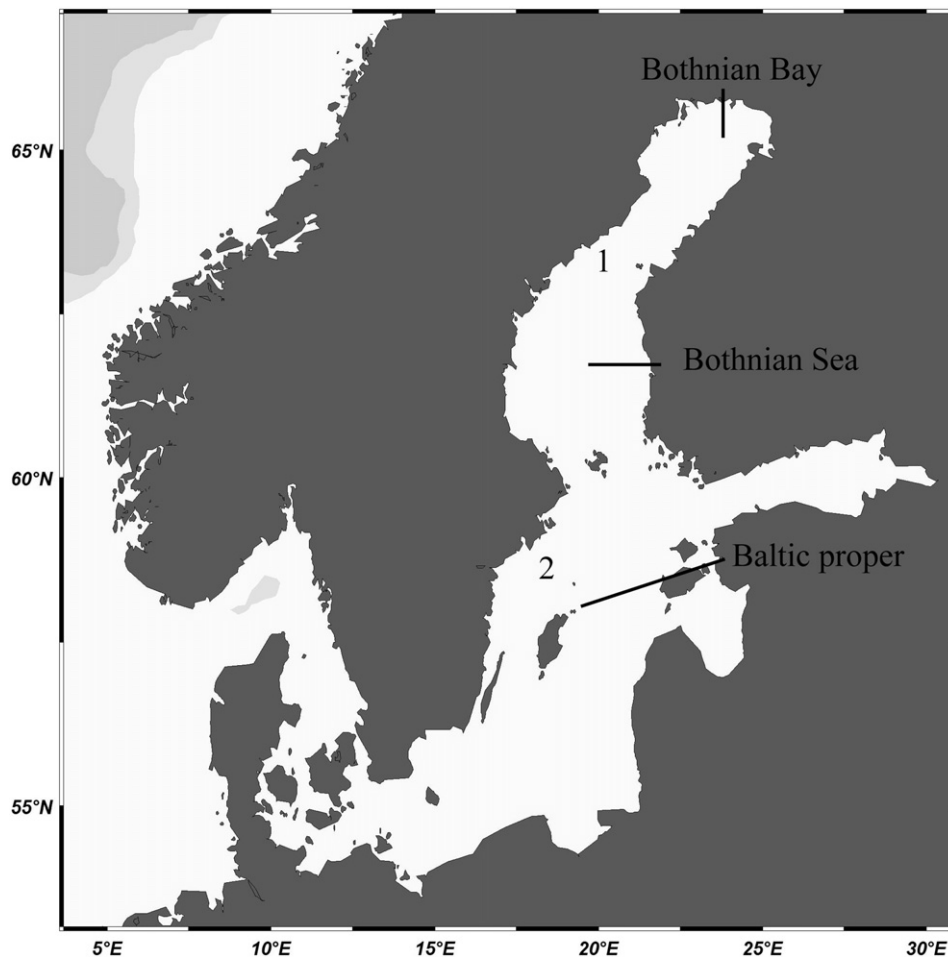


Fig. 1. Map of the Baltic Sea and its basins. 1=Bothnian Sea location for amphipod and sediment sampling in experiment 1. 2=Baltic proper location for sediment and amphipod sampling in experiment 1 and 2.

Table 1

Gravid female survival in Experiment 1 and 2 (males die after mating, thus no data is available)

Survival exp. 1.	Bothnian Sea sediment Bothnian Sea amphipods 19.7±5.1	Bothnian Sea sediment Baltic proper amphipods 21.25±5.2	Baltic proper sediment Bothnian Sea amphipods 18.7±3.8	Baltic proper sediment Baltic proper amphipods 17.6±9.9
Survival exp. 2	<i>Thalassiosira weissflogii</i>	<i>Anabaena</i> sp	<i>Rhodomonas salina</i>	<i>Thalassiosira weissflogii</i> and competition
	12.6±9.9	9.2±10.0	9.8±7.1	17.0±9.8

To each microcosm 50 adults is added and the sex ratio is expected to be 1:1 (Sundelin & Eriksson 2008, Sundelin et al., 2008). There were no significant differences in survival between treatments. Data represents mean values of 5 replicates±SD.

Several plausible hypotheses have been suggested as responsible for the drastic declines, including competition, new anthropogenic substances, hypoxia and predation. We argue that none of these have caused the Gulf of Bothnia decline. In the Great Lakes, competition for available food with the invasive species *Dreissena polymorpha* (zebra mussel) and *Dreissena buggia* (quagga mussel) has been suggested as possibly causing the decline of *Diporeia* sp (Nalepa et al., 2000; Nalepa et al., 2006; Watkins et al., 2007). In the Gulf of Bothnia the invading detritus feeding polychaete *Marenzelleria* spp (Bastrop and Blank, 2006) has been suggested to play the corresponding part. *Marenzelleria* spp. first occurred in the southern Baltic Sea in 1985 and has successfully migrated into Bothnian Bay. On Swedish monitoring stations in the Gulf of Bothnia, the increasing abundance of *Marenzelleria* spp. does not correlate to decreasing abundance of *M. affinis* (Karlsson and Leonardsson, 2005, monitoring data SHARK database SMHI, Sweden). Anthropogenic substances are unlikely since contaminant and biological effect monitoring in the Baltic do not show increasing trends (Bignert and Asplund, 2005; Sundelin et al., 2005, monitoring data, SHARK database SMHI, Sweden). Even if the bottom water oxygen concentration in the Bothnian Sea has decreased slightly during the last few years (Bjerkbæk-Lindberg and Gorringe, 2003) the oxygen concentrations are way beyond concentrations negatively affecting *M. affinis* (Eriksson Wiklund and Sundelin, 2001, 2004), which rules out hypoxia. Predation is also unlikely since the density of *M. affinis* main predator *Saduria entomon* has decreased simultaneously during the corresponding period (Leonardsson and Karlsson, 2003, monitoring data SHARK database SMHI, Sweden). Predation by fish is also unlikely since the populations of herring *Clupea harengus* in the Bothnian Sea have been stable since the early 1990ies (Ask and Westerberg, 2007).

In the years prior to the dramatic decline the highest river run-offs ever were observed (Alexandersson and Ehlert, 2004). The run-off transports large amount of alloctonous organic carbon, which reduces the light

conditions. The lowest extinction coefficient ever measured, was observed during the high run-off years (Alexandersson and Ehlert, 2004). This event could have contributed to the low phytoplankton bloom observed in the Bothnian Sea between 1998–2000 (Andersson and Wikner, 2004). The bacterial food web is favoured by a high freshwater run-off and is thereby fuelled by the alloctonous carbon (Sandberg et al., 2004), which might explain the higher ratio between bacterial and phytoplankton production observed during the high run-off years (Andersson and Wikner, 2004). A bacterial food web has many disadvantages for benthos compared to a phytoplankton based food web, i.e., a lower food web efficiency (Berglund et al., 2007) and a lower sedimentation.

In relation to these indices we hypothesize that the increased river outflow into the Gulf of Bothnia has resulted in events decreasing the availability of food resources and that the dramatic population decline is caused by malnutrition in terms of food quality and/or quantity. In this study we examine the impact of food and competition on amphipod reproductive outcome. In the first experiment we study effects of food quantity by incubating amphipods during oogenesis, mating and early embryo development in natural sediments differing in carbon content. In the second experiment we examine effects of food quality and competition, by feeding the amphipods (also during oogenesis, mating and early embryo development) with three different species of cultured micro algae, and by using the invasive species *Marenzelleria* spp. as competitor.

2. Material and Methods

2.1. Experimental set up

To determine the effect of food quantity in experiment 1 we exposed *M. affinis* from both Bothnian Sea and the Baltic proper (Fig. 1) to: 1) natural sediments from both their native sites, and 2) the Baltic proper amphipods to Bothnian Sea sediment and Bothnian Sea amphipods to

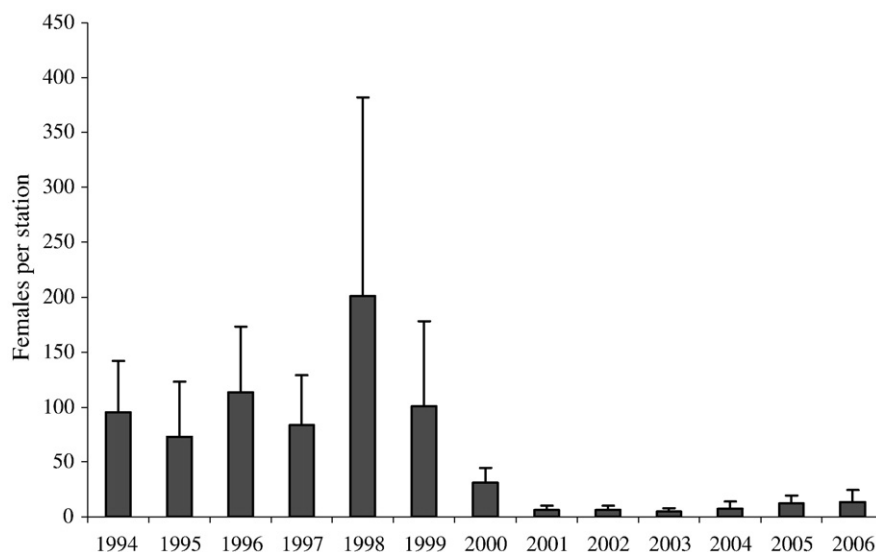


Fig. 2. The population decline in the Bothnian Sea here presented as the average number of gravid females of *Monoporeia affinis* collected in 20 Van Veen grabs from 4 coastal stations in the Swedish National monitoring Programme (5 grabs per station. Error bars represent SD).

Baltic proper sediment. Generally, the sediments in the Bothnian Sea contain less carbon than the sediments of the Baltic proper due to lower primary production in the Bothnian Sea. The sediment and amphipods from the Bothnian Sea were collected at 63° 25', 20° 07' at approximately 80 m depth and the sediment and amphipods from the Baltic proper were collected at 58° 49' N, 17° 38' at 30–40 m depth. The number of adult amphipods collected in the Bothnian Sea determined the size of the experiment. We used a microcosm system developed by Sundelin (1983). Briefly, the microcosm system consists of 2 L Ehrlenmeyer flasks (bottom area 227 cm²). The water inlet is placed below the outlet to produce a natural flow of water and to minimise gradients. The flasks are wrapped in green plastic to simulate the wavelength *in situ* and a timer regulated light to natural day length. To each microcosm we added 300 ml of Bothnian Sea or Baltic proper sediment and 50 amphipods of either origin. Each combination included 5 replicates. The experiment was conducted during four months, from September to January, covering oogenesis, mating and embryonic development.

In experiment 2 we used the same microcosm system and abundance per microcosm as in experiment 1. To all microcosms clean acid washed sand, containing no carbon, was added to a depth of 1.5 cm. Food quality was examined by adding algal suspensions of three different species cultured algae, the diatom *Thalassiosira weissflogii*, the cyanobacteria *Anabaena* spp and the cryptophyceae *Rhodomonas salina*. To address competition we made an additional series to which we added 25 individuals per microcosm (correspond to 1100 individuals m⁻²) of *Marenzelleria* spp. of 2–3 cm length combined with the addition of *T. weissflogii*, which was supposed to be the best food to *M. affinis*. We used 5 replicates of each series making a total of four series.

The algae were cultured semi-statically in standard culturing mediums adapted for each species and were harvested every 3–4 day. The cultured algae were allowed to settle and were washed with natural brackish water before carbon content was determined. The carbon content of the algal suspensions was determined by measuring cell density and size and applying standard equations (Hasle and Fryxell, 1977; Hill and Wetherbee, 1989; Menden-Deur and Lessard, 2000). Algae corresponding to 41 mg fresh carbon of each algae species were added three times during the exposure period to respective microcosm.

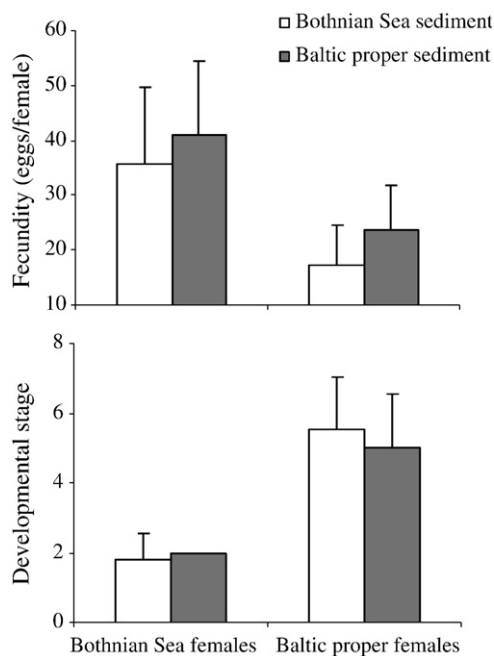


Fig. 3. Fecundity of *Monoporeia affinis* (eggs/female) and developmental stage of offspring from females off Bothnian Sea and the Baltic proper incubated in their native sediments or moved to the sediment from the other basin during gonad development, mating and early embryo development. Data is given as mean values of 5 replicates + SD.

Table 2

Experiment 1. Fatty acid content (μg/mg dw) of females sampled in the Bothnian Sea and the Baltic proper

Fatty acids	Female Baltic proper	Female Bothnian Sea
14:0	11.23	12.63
15:0	2.29	1.23
16:0	37.34	31.63
17:0	1.56	0.86
18:0	3.66	2.98
19:0	0.46	0.38
20:0	0.64	0.86
22:0	0.07	0.10
Σ Saturated	57.26	50.68
Σ Branched*	3.04	1.86
16:1n-7	64.79	56.64
17:1n-8	2.52	1.03
18:1n-9	29.20	40.06
18:1n-7	7.49	5.21
20:1n-9	0.82	0.87
20:1n-7	1.10	0.63
22:1n-11	0.07	0.01
22:1n-9	0.92	0.60
Σ Monounsaturated	106.89	105.05
16:3n-3	3.76	2.26
16:4n-3	2.25	1.64
18:2n-6	4.61	4.54
18:3n-6	0.05	0.14
18:3n-3	2.41	1.31
18:4n-3	7.52	5.83
20:2n-6	0.53	0.43
20:4n-6	5.05	3.04
20:3n-3	0.24	0.14
20:4n-3	0.71	0.64
20:5n-3	24.44	22.86
22:4n-6	0.38	0.35
22:5n-6	0.70	0.36
22:5n-3	0.58	0.44
22:6n-3	12.13	14.70
Σ Polyunsaturated	65.81	59.06
Σ HUFA	53.76	49.85
Σ Unsaturated	172.71	164.11
Σ (n-3)	54.04	49.82
Σ (n-6)	11.31	8.86
(n-3)/(n-6)	4.78	5.62
DHA/EPA	0.50	0.64
Σ Total FAME	232.54	216.27

* (Iso 14:0, Anteiso 14:0, Iso 16:0 and Anteiso 16:0).

To allow the algae to settle the flow through system was turned off for 24 h after the algal additions. Amphipods were exposed for three months, during the oogenesis, mating and early embryogenesis and the experiment ended in December. Yet, since these experiments were done in a short period of time, delayed and cumulative effects may be expected. Due to the low amphipod density at coastal stations in the Bothnian Sea the amphipods were collected in the Baltic proper at the same site as in experiment 1.

2.2. Embryo analyses

At experiment termination of both experiments embryos were carefully removed from the living female and analysed, using a stereomicroscope (80×magnification, Leica Z12), with respect to developmental stage and embryo aberrations. In this paper we have used the 9 developmental stages described in Sundelin and Eriksson (1998), which describes stages from single cell eggs to juveniles ready to hatch. The classified aberrations are malformed embryos, unfertilised/undeveloped eggs (henceforth called undifferentiated eggs), dead eggs and dead broods (see Sundelin and Eriksson, 1998; Sundelin et al., 2008a). Embryos are classified as malformed when they show various degrees of deformities, of eyes, legs, midgut, and membrane dysfunction, which means that lipids leak out through the membranes or water content of

embryos increases causing oedema. Undifferentiated eggs are either unfertilised or have ceased in development, which means that they are undeveloped as compared to other eggs in the brood. All the described embryo aberrations lead to mortality.

2.3. Fatty acid and carbon analyses

After teratogenic examination eggs and embryos were stored in -80°C to eliminate the risk of oxidation of fatty acids (Goedkoop et al., 2000). The determination of fatty acid profile was based on the experimental procedure described by Rosa et al. (2005a,b). The fatty acid methyl esters were analysed in a Varian 3400 gas chromatograph, equipped with an auto-sampler and fitted with a flame ionisation detector. The separation was carried out with helium as carrier gas in a fused silica capillary column Chrompack CPSil/88 (50 m $\times$ 0.32 mm id), programmed from 180°C to 200°C at $4^{\circ}\text{C min}^{-1}$, held for 10 min at 200°C and heated to 210°C for 14.5 min, with a detector at 250°C . A split injector (100:1) at 250°C was used. Fatty acid methyl esters were identified by comparison of their retention time with those of chromatographic Sigma standards. Since the response factor of the FID depends mainly on the carbon number of compounds, the peaks of compounds, for which no standard was available, were calibrated relative to other compound of the same carbon number. Peak areas were determined using the Varian software and the fatty acid 21:0 was used as an internal standard. The following FAs were identified in amphipods and algae cultures (Table 1). Chlorophyll a in the sediment was analysed according to Murphy et al. (2005) and Porra et al. (1989) with the modification of using acetone as solvent. Sediment organic

and nitrogen content was analysed by the accredited laboratory at the Department of Systems Ecology, Stockholm University.

2.4. Statistical analyses

In both experiments differences in treatments between embryo variables were tested by two-way- analysis of variance. In experiment 1 sediment and amphipod origin were used as independent variables and in experiment 2 food and competition were used as independent variables. Correlations between fatty acids and embryo variables were examined by Pearson correlation test. Data regarding embryo aberrations was logit-transformed prior to the analyses. All statistical tests were conducted using SPSS statistical package.

3. Results

3.1. Experiment 1

The reduction of amphipods in the Bothnian Sea is here presented as the number of gravid females collected in a Swedish National Monitoring Programme (Fig. 2.) The carbon content of the Baltic proper and the Bothnian Sea sediments were 4.85% and 0.55%, respectively, while the nitrogen contents were 0.70% and 0.07%. The sediment chlorophyll a content in the Baltic proper was $16.7\text{ }\mu\text{g/g dw}$ and $2.18\text{ }\mu\text{g/g dw}$ in the Bothnian Sea. Female survival did not differ between treatments ($p>0.05$, Table 1). Amphipod fecundity was significantly affected both by sediment (Fig. 3, $p<0.05$) and amphipod origin (Fig. 3, $p<0.001$) but there were no significant cross effects. Amphipods from

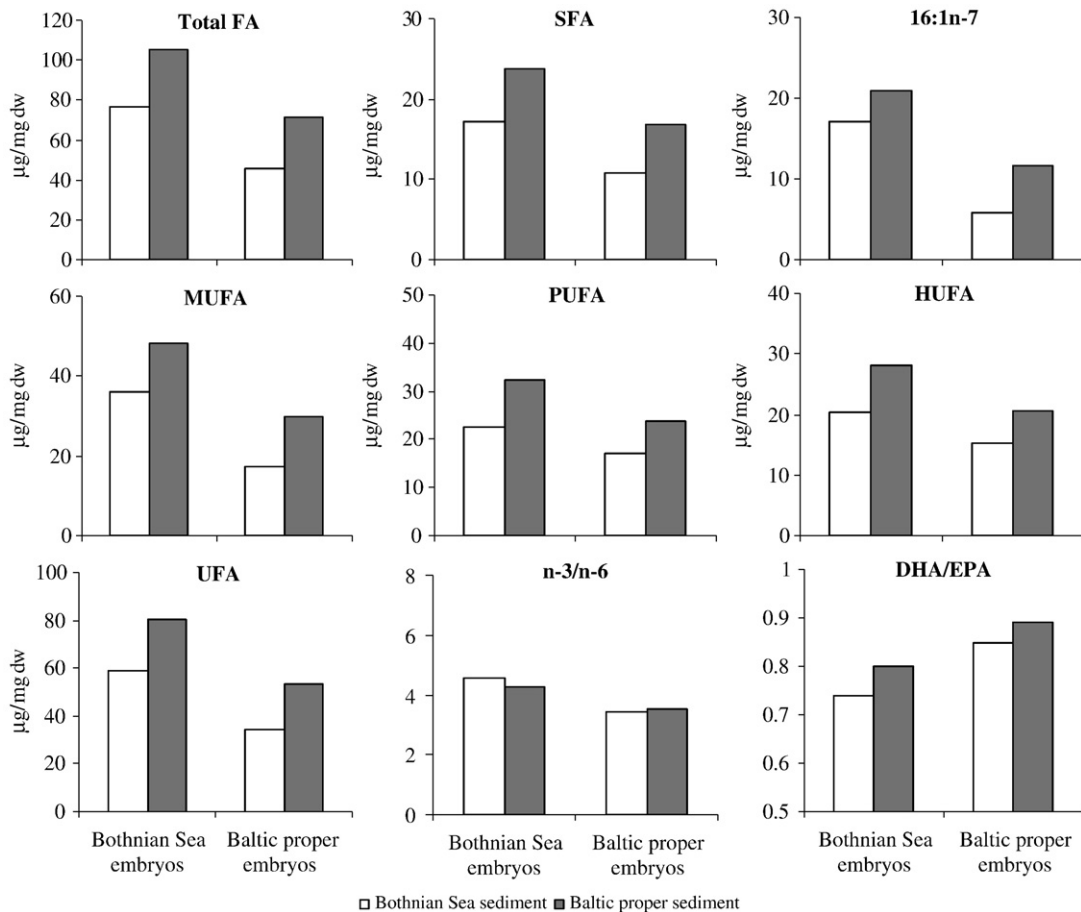


Fig. 4. Concentrations of total fatty acids, fatty acid classes and 16:1n-7 ($\mu\text{g/mg dw}$) in offspring from females reared in their native sediment or moved to the sediment from the other basin. Due to insufficient sample amount, offspring in each treatment were pooled without replicates. Abbreviations: total FA – total fatty acids; SFA – saturated fatty acids; MUFA – monounsaturated fatty acids; PUFA – polyunsaturated fatty acids; HUFA – highly unsaturated fatty acids; UFA – unsaturated fatty acids; EPA – 20:5n-3; DHA – 22:6n-3.

Table 3

Experiment 2. Fatty acid content ($\mu\text{g}/\text{mg}$ dw) of algal cultures added to each microcosms. Values are means of duplicate samples $\pm$ S.D

	<i>Thalassiosira weissflogii</i>	<i>Rhodomonas salina</i>	<i>Anabena</i> sp.
14:0	8.17 $\pm$ 0.24	7.24 $\pm$ 0.50	2.03 $\pm$ 0.01
15:0	0.84 $\pm$ 0.11	0.28 $\pm$ 0.01	0.29 $\pm$ 0.02
16:0	6.58 $\pm$ 0.51	5.31 $\pm$ 0.47	7.79 $\pm$ 0.24
17:0	0.09 $\pm$ 0.01	0.04 $\pm$ 0.05	0.11 $\pm$ 0.03
18:0	0.74 $\pm$ 0.05	0.79 $\pm$ 0.09	1.10 $\pm$ 0.08
19:0	0.08 $\pm$ 0.01	0.13 $\pm$ 0.02	0.02 $\pm$ 0.01
20:0	0.06 $\pm$ 0.04	0.01 $\pm$ 0.01	0.10 $\pm$ 0.01
22:0	0.08 $\pm$ 0.01	tr.	0.02 $\pm$ 0.03
Σ Saturated	16.65 $\pm$ 0.87	13.80 $\pm$ 1.16	11.46 $\pm$ 0.37
Σ Branched	1.88 $\pm$ 0.03	1.88 $\pm$ 0.15	2.26 $\pm$ 0.19
16:1n-7	9.25 $\pm$ 1.44	2.47 $\pm$ 0.27	9.32 $\pm$ 0.18
17:1n-8	0.15 $\pm$ 0.01	0.12 $\pm$ 0.02	0.32 $\pm$ 0.04
18:1n-9	1.43 $\pm$ 0.16	1.93 $\pm$ 0.29	4.06 $\pm$ 0.76
18:1n-7	0.66 $\pm$ 0.30	2.42 $\pm$ 0.18	1.96 $\pm$ 0.12
20:1n-9	tr.	0.17 $\pm$ 0.01	0.04 $\pm$ 0.06
20:1n-7	0.01 $\pm$ 0.01	0.02 $\pm$ 0.02	0.06 $\pm$ 0.03
22:1n-11	tr.	tr.	0.02 $\pm$ 0.02
22:1n-9	tr.	tr.	0.01 $\pm$ 0.02
Σ Monounsaturated	11.50 $\pm$ 1.32	7.12 $\pm$ 0.21	15.79 $\pm$ 0.87
16:3n-3	11.24 $\pm$ 1.74	0.71 $\pm$ 0.20	2.26 $\pm$ 0.16
16:4n-3	0.50 $\pm$ 0.02	0.59 $\pm$ 0.03	0.04 $\pm$ 0.00
18:2n-6	1.90 $\pm$ 0.29	2.07 $\pm$ 0.07	2.75 $\pm$ 0.02
18:3n-6	0.04 $\pm$ 0.00	tr.	0.14 $\pm$ 0.03
18:3n-3	2.44 $\pm$ 0.94	18.41 $\pm$ 1.58	3.65 $\pm$ 0.08
18:4n-3	4.41 $\pm$ 1.84	24.58 $\pm$ 2.51	0.42 $\pm$ 0.05
20:2n-6	0.77 $\pm$ 0.13	0.02 $\pm$ 0.02	0.11 $\pm$ 0.10
20:4n-6	0.20 $\pm$ 0.00	0.21 $\pm$ 0.07	0.11 $\pm$ 0.07
20:3n-3	0.01 $\pm$ 0.02	0.02 $\pm$ 0.03	0.02 $\pm$ 0.04
20:4n-3	0.18 $\pm$ 0.04	0.39 $\pm$ 0.05	0.05 $\pm$ 0.02
20:5n-3	12.22 $\pm$ 0.29	8.65 $\pm$ 0.81	3.05 $\pm$ 0.51
22:4n-6	0.32 $\pm$ 0.01	0.27 $\pm$ 0.39	0.16 $\pm$ 0.20
22:5n-6	0.02 $\pm$ 0.03	tr.	0.01 $\pm$ 0.02
22:5n-3	0.16 $\pm$ 0.09	0.11 $\pm$ 0.15	0.04 $\pm$ 0.05
22:6n-3	4.62 $\pm$ 0.54	6.30 $\pm$ 0.76	1.30 $\pm$ 0.41
Σ Polyunsaturated	39.10 $\pm$ 5.06	62.45 $\pm$ 5.51	14.13 $\pm$ 0.59
Σ HUFA	22.63 $\pm$ 4.29	41.10 $\pm$ 3.99	5.17 $\pm$ 0.93
Σ Unsaturated	50.60 $\pm$ 3.74	69.58 $\pm$ 5.72	29.91 $\pm$ 1.46
Σ (n-3)	35.78 $\pm$ 5.43	59.76 $\pm$ 5.71	10.83 $\pm$ 0.84
Σ (n-6)	3.24 $\pm$ 0.37	2.57 $\pm$ 0.22	3.27 $\pm$ 0.26
(n-3)/(n-6)	11.21 $\pm$ 2.97	23.47 $\pm$ 4.28	3.33 $\pm$ 0.52
DHA/EPA	0.38 $\pm$ 0.05	0.73 $\pm$ 0.02	0.42 $\pm$ 0.06
Σ Total FAME	69.05 $\pm$ 2.83	85.12 $\pm$ 7.01	43.61 $\pm$ 2.02

* (Iso 14:0, Anteiso 14:0, Iso 16:0 and Anteiso 16:0); tr – trace.

the Bothnian Sea increased their fecundity by 13% when moved to the richer Baltic proper sediment, while fecundity in amphipods from the Baltic proper was reduced by 25% when moved to the poorer Bothnian Sea sediment. Regarding embryogenesis, offspring originating from the Baltic proper females reached a higher developmental stage than those originating from Bothnian Sea females (Fig. 3, $p < 0.001$). Exposure to the Bothnian Sea or Baltic proper sediments did not affect developmental stage ($p > 0.05$) and no cross effects between amphipod and sediment origin were found. Also, there were no differences in the occurrence of malformed, undifferentiated or dead eggs between amphipods or sediments from the Baltic proper or the Bothnian Sea.

The total fatty acid (FA) content of the females from the Baltic proper and Bothnian Sea were 232 and 216 $\mu\text{g}/\text{mg}$ dry weight (dw), respectively (Table 2). The higher FA content found in the former group was a result of higher concentrations of saturated (SFA) and polyunsaturated (PUFA) fatty acids. The monounsaturated (MUFA) fraction was quite similar (106.9 and 105.1 $\mu\text{g}/\text{mg}$ dw, respectively). After 4 months of exposure to the two sediments, the offspring of the respective females were affected in an opposite matter. Offspring of females from Bothnian Sea and exposed in Baltic proper sediment showed the highest FA concentrations, while offspring of females from Baltic proper and exposed in Bothnian Sea sediment revealed the lowest FA values (Fig. 4). Additionally, females incubated in their

native sediments produced offspring with similar FA levels. Fecundity was positively correlated to total FAs in the offspring and the main classes of FA ($p < 0.001$). On the other hand, developmental stage was negatively correlated to the total FA content of the offspring ($p < 0.001$) and the main FA classes. We did not measure FA concentration in the spent females since they could get damaged while removing the eggs. No correlation between sediment food characteristics and the incidence of malformed embryos was observed ($p > 0.05$; data not shown).

3.2. Experiment 2

The fatty acid profile of the three algal cultures is shown in Table 2. The total FA content was highest in the *R. salina* culture (85.12 $\mu\text{g}/\text{mg}$ dw) followed by *T. weissflogii* (69.05 $\mu\text{g}/\text{mg}$ dw) (Table 3). The highest amount of SFA was found in *T. weissflogii* (16.65 $\mu\text{g}/\text{mg}$ dw), while MUFA levels were higher in the *Anabaena* sp. culture (15.79 $\mu\text{g}/\text{mg}$ dw) followed by *T. weissflogii* (11.50 $\mu\text{g}/\text{mg}$ dw). The 16:1n-7 (considered as a marker of diatoms) was found in equal concentrations in both *T. weissflogii* and *Anabaena* sp. The levels of the other major FA classes, namely PUFA and HUFA, were highest in the *R. salina* culture.

Table 4

Experiment 2. Fatty acid content ($\mu\text{g}/\text{mg}$ dw) of the offspring whose parent was reared on these algal diets

Fatty acids	<i>Thalassiosira weissflogii</i>	<i>Rhodomonas salina</i>	<i>Anabena</i> sp.	<i>T. weissflogii</i> and competition
14:0	4.31	5.29	3.55	5.44
15:0	1.16	0.91	0.53	1.49
16:0	23.15	26.34	25.68	27.12
17:0	0.80	1.00	1.04	0.91
18:0	2.06	2.47	2.66	2.25
19:0	0.20	0.33	0.19	0.25
20:0	tr.	0.05	0.06	0.06
22:0	tr.	tr.	0.07	0.11
Σ Saturated	31.68	36.41	33.78	37.63
Σ Branched	1.23	1.26	1.29	1.37
16:1n-7	18.29	17.36	16.91	21.70
17:1n-8	0.98	1.01	0.99	1.12
18:1n-9	32.60	42.58	43.61	37.64
18:1n-7	4.61	6.03	5.77	5.60
20:1n-9	0.97	1.02	1.17	0.94
20:1n-7	1.01	1.23	1.55	1.39
22:1n-11	0.20	0.25	0.22	0.24
22:1n-9	0.21	0.23	0.24	0.25
Σ Monounsaturated	58.87	69.71	70.46	68.89
16:3n-3	2.01	0.92	0.56	2.40
16:4n-3	0.46	0.65	0.55	0.35
18:2n-6	4.66	6.86	6.56	5.74
18:3n-6	0.19	0.17	0.17	0.21
18:3n-3	3.36	7.12	5.94	4.04
18:4n-3	3.35	5.48	3.73	4.06
20:2n-6	0.53	0.73	0.76	0.56
20:4n-6	1.32	1.51	1.51	1.33
20:3n-3	0.45	0.57	0.54	0.51
20:4n-3	0.73	0.92	0.78	0.82
20:5n-3	23.45	23.80	22.40	25.25
22:4n-6	0.63	0.51	0.71	0.74
22:5n-6	0.00	0.31	0.33	0.28
22:5n-3	0.75	0.82	0.82	0.78
22:6n-3	15.85	17.62	18.02	16.74
Σ Polyunsaturated	57.92	68.32	63.58	64.03
Σ HUFA	46.53	51.62	48.86	50.33
Σ Unsaturated	116.80	138.03	134.04	132.92
Σ (n-3)	50.41	57.90	53.35	54.94
Σ (n-6)	7.32	10.09	10.04	8.85
(n-3)/(n-6)	6.89	5.74	5.31	6.21
DHA/EPA	0.68	0.74	0.80	0.66
Σ Total FAME	149.50	175.36	168.93	171.67

* (Iso 14:0, Anteiso 14:0, Iso 16:0 and Anteiso 16:0); tr – trace.

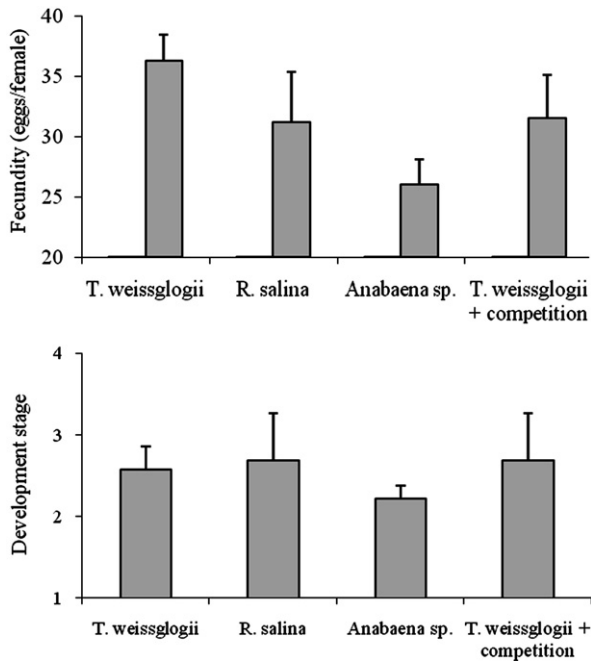


Fig. 5. Fecundity (eggs/female) of *Monoporeia affinis* and developmental stage of offspring from females fed three different diets and one diet in combination with competition. The three algal additions were *Thalassiosira weissflogii*, *Rhodomonas salina* and *Anabaena* sp. The fourth treatment was *T. weissflogii* combined with competition from the polychaete *Marenzelleria* spp. Data represent mean values $\pm$ SD.

After feeding the amphipod females with the algal diets, UFA levels were the main contributor to the differences in FA content in the developing embryos (Table 4). The highest total FA content was observed in the offspring of females fed on *R. salina*. Amphipods fed on *T. weissflogii* produced offspring with higher n-3/ n-6 ratio (6.89). Competition of food with the invasive species *Marenzelleria* spp. did not reduce the FA content in the embryos of *M. affinis*. There was no correlation between developmental stage and FA content ($p > 0.05$). Fecundity was negatively correlated to MUFA content ($p < 0.05$) and the n-3/ n-6 ratio ($p < 0.01$).

The fecundity was lower in the *Anabaena* treatment compared to the *Thalassiosira* and *Rhodomonas* treatments ($p < 0.05$) while no effect from competition was observed. Neither food quality nor competition had significant effect on amphipod survival ($p > 0.05$; Fig. 5, Table 1) even if less gravid females was found compared to experiment 1. Also, the developmental stage was not affected ($p > 0.05$) since all embryos had reached approximately the same developmental stage, which varied between 2.2 and 2.7 (Fig. 5). We did not observe any significant relationship between food quality and competition on the incidence of malformed, dead or undifferentiated embryos ($p > 0.05$; data not shown).

4. Discussion

4.1. Significance of food content

Sediment properties, in terms of OC content and Chl a, affected the reproduction of *Monoporeia affinis*. When females in experiment 1 were transferred from their native sediment to richer (Baltic proper sediment) or poorer (Bothnian Sea) sediments, the impact of the altered sediment characteristics was clearly demonstrated. Females reduced their fecundity by 25% when moved from the richer Baltic proper sediment to the Bothnian Sea sediment and the fatty acid content in the offspring was also reduced compared to offspring from females incubated in their native Baltic proper sediment. The opposite pattern occurred when Bothnian Sea females were incubated in the richer Baltic proper sediments in terms of increased fecundity and higher fatty acid content in the developing embryos. At experimental start, females from both

origins have approximately the same FA-concentration, as do offspring from females reared in their native sediment, indicating that females are adapted to the nutritional status of their habitat. Offspring from the Baltic proper had reached a significantly higher developmental stage and had a lower FA content irrespective of treatment compared to the offspring of Bothnian Sea. This difference between FA concentrations could be explained by the differences in developmental stage. The developing embryos catabolise their stored energy reserves, mainly MUFA during embryogenesis (Morais et al., 2002; Rosa et al., 2005a, 2007), which leads to a continuous reduction in embryo's lipid content.

Irrespective of treatment the females collected in the Bothnian Sea produced more eggs than those from the Baltic proper. Regarding the lower nutritional content of the Bothnian Sea sediment this might appear contradictory, but fecundity is closely related to size (Cederwall, 1977; Leonardsson et al., 1988; Glazier, 2000; Subida et al., 2005). Before the decline, it seems as growth was food limited in the Bothnian Sea, but after the decline the size has increased (Karlsson and Leonardsson, 2005), indicating sufficient amount of food resources also in the poorer Bothnian Sea sediments. A second factor that influences female size is the length of *M. affinis* life cycle, which increases northwards. The average life span is 3 to 4 years in the Gulf of Bothnia, while Baltic proper populations of the same depth have an average life span of 2 years (Leonardsson et al., 1988 and own observations). A delayed maturation in an environment with sufficient food resources might result in sexually mature females with larger size and higher fecundity. However, this difference in size and fecundity of the populations do not influence the outcome of the experiment since we do not make any direct comparisons between females of different origin. The increase in female size and fecundity is presently leading to a population increase on many Bothnian Sea stations, but since the increase in several cases is starting from a very low level and the life cycle in the Bothnian Sea is 3–4 years the predicted recovery time is as long as 12 years on many coastal stations (Karlsson and Leonardsson, 2005).

Our result clearly demonstrates that a disturbance in the food supply during oogenesis alters the reproductive outcome. Females adjust the investment in offspring in order to enhance their chance of survival during the low food winter months, by adjusting the number of eggs and not the size of the eggs, (Sundelin and Eriksson, 1998). This implies that these amphipods make a trade-off between somatic conditions and reproductive investment. These results are in line with e.g. Elkin and Reid (2005) whom implies that the ability to feed at the breeding site affected the reproductive potential in beetles and is fundamental in a life-history perspective (Williams, 1966; Rosa and Nunes, 2002, 2003). Our results contradict the results of Aljetjawi and Leonardsson (2003) who suggested that females lacked the ability to adjust the offspring investment. A lack of such ability would severely affect female survival due to the long brooding period. In contrast to many iteroparous species being able to interrupt reproduction during starvation (Stelzer, 2001), the semelparous *M. affinis* lacks this ability to postpone the reproduction to a forthcoming year even at a very early stage of gonad maturation. During impaired environmental conditions dead and undeveloped oocytes will be lost when mature oocytes are transferred into the marsupium before fertilisation. Food shortage during late oogenesis could result in unsuccessful fertilisation and unfertilised females die during spring similarly to spent females when they have released their offspring (Sundelin et al., 2008b). During severe starvation it is possible that ovigerous females are unable to maintain somatic growth and die during autumn before fertilisation since a high starvation sensitivity of the adult generation has been demonstrated (Aljetjawi and Leonardsson, 2002, 2003). In conclusion, *M. affinis* could adapt to food shortage in early oogenesis by reducing the offspring investment, while disturbance in later oogenesis might imply that fertilisation totally fails and one generation will be lost. In a life history perspective, a semelparous reproductive strategy combined with no ability to utilize food resources already allocated into offspring investment would be detrimental.

4.2. Significance of food quality and competition

In experiment 2 the adult generation was offered the same amount of carbon from three food sources varying in nutritional value. Diatoms normally dominating the spring bloom have been regarded as high quality food resources compared to bacteria and other species of algae (Müller-Navarra, 1995; Goedkoop et al., 2000) and higher fecundity was expected in the diatom series. However, in the present study, there was only weak evidence of a beneficial effect of food quality on amphipod reproduction, development and nutritional condition of the embryos. The fecundity was lower in the *Anabaena* treatment compared to the *Thalassiosira* and *Rhodomonas* treatments while a total FA content in the *R. salina* culture twice the concentration in the *Anabaena* sp. culture (Table 2), was not reflected in the offspring (Table 3). Our result corresponds with results on field collected *Calanus finmarchius* and *C. helgolandicus* (Jonasdottir et al., 2005). They found that egg production rate had no positive relationship to any fatty acid but hatching success was correlated to the FA ratio 22:6n-3/20:5n-3. Also Boersma and Stelzer (2000) found very low response to fatty acid addition in copepods and selective feeding cladocerans. It is possible that female *M. affinis* can accumulate sufficient amounts of high quality essential FAs, specifically allocated for oocyte production from spring and summer blooms, while their own metabolism is maintained on low quality food. The main diatom bloom in the Baltic occurs in spring, . during late summer or early autumn, blooms are dominated by other taxonomic groups than diatoms. This means that these amphipods cannot rely on diatoms during their reproductive period. We cannot exclude the possibility that longer experimental duration covering a larger part of the embryo development, would result in effects due to food quality but it seems unlikely in this case, since the developing embryos use their own stored resources. On the other hand, if we had started earlier in the female life cycle, differences in embryo FA content and composition depending on parental food source might be plausible. In addition, the role of diatoms in aquatic food webs has been questioned in recent years due to their teratogenic effects on copepods caused by aldehydes (Miralto et al., 1999; Casotti et al., 2005). It changed the established concept of energy flow from diatoms to fish by means of copepods. This is an ongoing debate since the ecological significance of the laboratory experiments has been questioned (Jones and Flynn, 2005). In addition, neurotoxins in the form of domoic acid have been observed in some taxonomic groups of diatoms (see Costa et al., 2005a,b). As already mentioned, competition by the invasive species *Marenzelleria* spp. has been suggested causing the disappearance of *M. affinis* from the Bothnian Sea. Previous studies have reported that the presence of *Marenzelleria* spp. affected growth of *M. affinis* at moderate densities which was attributed to food competition rather than interference competition (Kotta et al., 2001; Kotta and Ólafsson, 2003). Before the amphipod decline, the abundance of *Marenzelleria* spp. was less than 10 ind /m² in large areas of the Bothnian Bay, but after the decline they increased dramatically in many areas (Karlsson and Leonardsson, 2004). The density of *Marenzelleria* spp used in this study, which is in comparison with polychaete abundance after the amphipod decline in the Bothnian Sea, did not affect survival or reproductive outcome and the density used was only slightly lower than densities that affect the distribution patterns of *M. affinis* in other laboratory experiments (Neideman et al., 2003). Although we have not fully addressed competition in our experiment, we can relate our results to monitoring observations, which give indications that at moderate densities both species can co-exist as long as they have access to sufficient food supply. Thus, it is unlikely that the low densities of *Marenzelleria* spp. observed before the amphipod decline affected amphipod reproduction. Monitoring data also shows that after the amphipod populations declined, *Marenzelleria* increased drastically, indicating a strong colonisation capability (Karlsson Leonardsson, 2005). It is then plausible that the increased abundance of *Marenzelleria* will increase food competition and further slow down the slow recovery process of *M. affinis*, which has

started in offshore areas (Monitoring data SHARK database, SMHI Sweden). It is also unknown if the population decline have caused any changes in the previously observed mortality patterns in respective year-class (Karlsson and Leonardsson, 2005).

According to previous observations (Sundelin and Eriksson, 1998; Sundelin et al., 2008a,b) the sex ratio is normally 1:1. However, we found less gravid females and higher variability between replicates in experiment 2 than in experiment 1. This might indicate that male and female reproductive stages were not completely synchronous in experiment 2. During sexual maturation males and females undergo a series of moulting stages. Mature males are recognized by their long antennae with aesthetascs facilitating recognition of females' chemical cues (McCurdy et al., 2000). In the related amphipod *Corophium volutator*, it has been shown that timing is of greatest importance in order to achieve reproductive success since females can be fertilised only a few days after the last molt when oocytes are healthy (McCurdy et al., 2000; Krång and Baden, 2004). If no sexually mature males are available the female oocytes will degenerate. When performing laboratory experiments in microcosms, the size of the microcosm is a trade-off between relevance and replication. A higher density of amphipods or larger microcosms, giving the mature females a larger selection of mature males would likely have increased the number of fertilised females.

4.3. The amphipod decline - interactions between climate change and primary production?

According to climate models increased temperature will lead to increased precipitation in northern Europe (Meier, 2006). The population declines in the Bothnian Sea coincided with several extreme events possibly related to climate change such as the highest run-offs into the Gulf of Bothnia ever measured in 1998 and 2000 (Alexandersson and Ehlert, 2004). This resulted in a reduced phytoplankton production and a higher ratio between bacteria production and phytoplankton production (Andersson and Wikner, 2004). If this process continues it can result in a shift from a phytoplankton based primary production towards a bacteria based, which leads to more trophic levels and a lower food web efficiency (Berglund et al., 2007). The increased amounts of freshwater could also result in an altered phytoplankton community created by a shallower upper mixed layer, terminating diatom growth earlier due to less available nutrients. Dinoflagellates will replace the diatoms earlier, since they can migrate into deeper nutrient-rich layers (Olli et al., 1998). A decreased diatom bloom may result in declining amounts of food resources in settling matter, as dinoflagellates to a large extent decompose in the water mass or settle as resting eggs non-digestible to benthic organisms (Heiskanen, 1993). In the Baltic proper a shift, from diatoms to dinoflagellates as dominating taxa, in the spring phytoplankton bloom has been recorded (Wasmund et al., 1998). Attempts have been made in the Great Lakes in North America, both experimentally and from field data, to establish relationships between food limitations and the continued disappearance of *Diporeia* sp (Nalepa et al., 2006; Watkins et al., 2007). Their studies could neither refute nor confirm this hypothesis. Even if the results of experiment 2 showed that reproduction was not favoured by diatoms, this group has a great advantage due to its sedimentation properties both compared to other algae groups and especially compared to a bacteria based food web. An altered primary production regime caused by increased run-off could severely reduce carbon input to benthos, drastically changing the conditions to a benthic community, which was considered to be food limited (Karlsson and Leonardsson, 2005). Our results showed that females seem to be adapted to a nutrient content in an area. In our experiment a period of altered conditions during one month before mating significantly affected fecundity and longer periods of starvation. If a sudden event affects food concentrations during reproduction, their accumulated lipids is not sufficient to

maintain egg production and their own metabolism, as the adult generation have demonstrated high mortality during starvation (Aljetlawi and Leonardsson, 2002, 2003). Since semelparity is considered as the optimal strategy in a constant environment (Gurney and Middleton, 1996) a higher variability in food supply caused by an increased incidence of extreme events could therefore have large implications for semelparous organisms like *M. affinis*. We suggest that food shortage has contributed to the population declines but it is not proven to be the single cause. A secondary effect of malnutrition could be increased susceptibility to other types of stressors such as contaminants, parasites etc. This is the first in a series of papers in which we examine the amphipod decline in the Gulf of Bothnia and the factors that might have caused it.

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